Reasoning paths and path energy. A reasoning path p from a source step s to a target step t is a sequence of edges

$$p = (b_1, \ldots, b_L), \quad b_\ell = (u_\ell \to u_{\ell+1}), \quad u_1 = s, \quad u_{L+1} = t. \tag{29}$$

Define the path energy as the additive cost

$$\mathcal{E}(p) \;=\; \sum_{\ell=1}^{L} E_{b_\ell}. \tag{30}$$

Soft-min (effective) energy over all paths. Let $\mathcal{P}(s \to t)$ denote the set of all paths from s to t in G. Following the standard log-sum-exp (soft-min) construction, define the effective energy

$$\mathcal{E}^\star(s \to t) \;=\; -\log \sum_{p \in \mathcal{P}(s \to t)} \exp\big(-\mathcal{E}(p)\big). \tag{31}$$

Equivalently, each path receives an (unnormalized) Gibbs weight $\exp(-\mathcal{E}(p))$, so lower-energy paths contribute more strongly to the aggregate dependence from s to t.

Low-energy edges define effective constraints. Consider two paths p and p' from s to t of equal length L. Assume they are identical except at a single position ℓ^*, where p uses edge b_{ℓ^*} and p' uses b'_{ℓ^*}. If $E_{b_{\ell^*}} \le E_{b'_{\ell^*}} - \delta$ for some $\delta > 0$, then

$$\frac{\exp\big(-\mathcal{E}(p)\big)}{\exp\big(-\mathcal{E}(p')\big)} = \exp\big(\mathcal{E}(p') - \mathcal{E}(p)\big) \;\ge\; \exp(\delta). \tag{32}$$

Thus, holding all other edges fixed, replacing a higher-energy edge by a lower-energy edge amplifies the relative influence of that path by at least a factor $\exp(\delta)$ in the attention-induced dependency distribution.

Proof. Because p and p' differ only at ℓ^*,

$$\mathcal{E}(p) - \mathcal{E}(p') = E_{b_{\ell^*}} - E_{b'_{\ell^*}}. \tag{33}$$

The assumption $E_{b_{\ell^*}} \le E_{b'_{\ell^*}} - \delta$ implies

$$\mathcal{E}(p) \le \mathcal{E}(p') - \delta \quad \Rightarrow \quad \mathcal{E}(p') - \mathcal{E}(p) \ge \delta. \tag{34}$$

Exponentiating yields

$$\frac{\exp\big(-\mathcal{E}(p)\big)}{\exp\big(-\mathcal{E}(p')\big)} = \exp\big(\mathcal{E}(p') - \mathcal{E}(p)\big) \ge \exp(\delta), \tag{35}$$

which proves the claim.

Analysis. Proposition C.1.4 shows that low-energy edges act as effective constraints: they bias the model toward reusing specific multi-step dependency patterns, because any path that swaps in a lower-energy edge gains multiplicative weight in the induced path distribution. If edges labeled as Deep Reasoning consistently exhibit lower energies than alternative behaviors, then multi-hop dependencies that rely more heavily on Deep Reasoning edges will dominate the effective aggregation from premises to conclusions, stabilizing long chain-of-thought structure without invoking an external chemical analogy.

C.1.5 Long CoT process is looking for a more stable reasoning structure.

Generally speaking, in chemistry, reactions tend to synthesize more stable, lower-energy compounds. Analogously, we hypothesize that the **Long CoT learning process** seeks a stable reasoning configuration that minimizes attention energy. We now formalize this intuition under mild ergodicity and bounded-energy assumptions.

Step 1. Stationary behavior frequencies. Assume that the reasoning behavior sequence $(s_t)_{t\geq 1}$ forms an irreducible, aperiodic, time-homogeneous Markov chain with transition matrix P and stationary distribution π satisfying $P^\top \pi = \pi$. By the ergodic theorem for finite-state Markov chains, the empirical frequency of each behavior converges almost surely to its stationary probability[§]:

$$\frac{1}{T-1}\sum_{t=1}^{T-1} \mathbf{1}[s_t = b] \xrightarrow[T\to\infty]{a.s.} \pi_b, \quad \forall b \in \mathcal{B}. \tag{36}$$

Step 2. Decomposition of the time-averaged energy. Let E_t denote the *bond-level attention energy* associated with transition $e_t = (u_t \to u_{t+1})$, as defined in Appendix C.4. The trajectory-level average energy can be decomposed by behavior type as

$$\widehat{E}_T = \sum_{b\in\mathcal{B}} \underbrace{\left(\frac{1}{T-1}\sum_{t=1}^{T-1}\mathbf{1}[s_t=b]\right)}_{\text{Empirical frequency of behavior } b} \cdot \underbrace{\left(\frac{\sum_{t=1}^{T-1} E_t\,\mathbf{1}[s_t=b]}{\sum_{t=1}^{T-1}\mathbf{1}[s_t=b]}\right)}_{\text{Mean energy conditional on } s_t=b}. \tag{37}$$

This simply states that the global average equals the weighted sum of per-behavior averages.

Step 3. Conditional convergence of energy averages. Assume that $\mathbb{E}[|E_t|] < \infty$ and, conditional on $s_t = b$, the distribution of E_t depends on the past only through s_t (i.e., $E_t \perp (s_{<t}, E_{<t}) \mid s_t$). Under this mild conditional-independence assumption, the law of large numbers within each behavior implies that the inner mean converges to the conditional expectation

$$\frac{1}{N_b(T)}\sum_{t:\, s_t=b} E_t \;\to\; \mu_b = \mathbb{E}[E_t \mid s_t = b]. \tag{38}$$

Combining this with the ergodic frequency limit from Step 1 yields

$$\widehat{E}_T \xrightarrow[T\to\infty]{a.s.} \sum_{b\in\mathcal{B}} \pi_b \mu_b, \tag{39}$$

thus establishing the ergodic low-energy equilibrium.

Step 4. Exponential routing preference from Gibbs attention. At reasoning step $t+1$, consider a fixed query token i selecting among candidate targets $\mathcal{S}_t$. The model samples target $s_j \in \mathcal{S}_t$ according to the standard Boltzmann (softmax) distribution:

$$\Pr(\mathcal{S}_t = s_j \mid i) = \frac{\exp(-E_{ij})}{\sum_{s_\ell\in\mathcal{S}_t}\exp(-E_{i\ell})}. \tag{40}$$

Assume a bounded-deviation condition: for any behavior type b, the corresponding attention energies lie within an interval $[\mu_b - \Delta, \mu_b + \Delta]$. Then for any two behavior classes b, c co-occurring in $\mathcal{S}_t$,

$$\frac{\Pr(\mathcal{S}_t \in b \mid i)}{\Pr(\mathcal{S}_t \in c \mid i)} \geq \exp\big(-(\mu_b+\Delta) + (\mu_c - \Delta)\big) = \exp\big((\mu_c - \mu_b) - 2\Delta\big). \tag{41}$$

Intuitively, lower-energy behaviors are exponentially favored in the routing distribution. If Deep/Reflection behaviors exhibit mean energies $\mu_{\mathcal{D}}, \mu_{\mathcal{R}}$ separated from Exploration energy $\mu_{\mathcal{E}}$ by a margin $\gamma > 0$, and $2\Delta < \gamma$, then these lower-energy transitions will dominate with exponential advantage $\exp(\gamma - 2\Delta)$.

Therefore, the model's attention mechanism inherently biases it toward stable, low-energy Long CoT reasoning structures.

[§]Note: The energy at t step is related to the previous logical node (node), but has nothing to do with the logical step (edge).

C.2 Stable Reasoning Bond Distribution in Long CoT

Reasoning Chain Generation To analyze structural properties across domains, we utilize OpenThoughts-3 [24], a mixed-domain corpus comprising math, code, and scientific reasoning tasks typical of QwQ-style distillation pipelines. We prompted DeepSeek-R1-671B, OpenAI-o1, and QwQ-32B with original queries to generate Long CoT traces. Generation employed a maximum token length of {16,384, 32,768}, a decoding temperature $T \in [0, 1]$, top-$p = 0.95$, and standard repetition penalties, without additional sampling constraints.

Bond Type Annotation Following Chen et al. [19], Golovneva et al. [34], we segmented reasoning traces into step-level units using standard delimiters (e.g., "\n", "\n\n", ". "). We then prompted Qwen2.5-32B-Instruct to classify the logical bond for each edge between consecutive steps (h_{t-1}, h_t) into one of three types. Validation on a 200-example subset against human annotation yielded a macro-F1 score exceeding 0.85, confirming the reliability of the automated labeling applied to the full corpus.

Bond Type Annotation Prompt

You are an expert annotator. Classify the CURRENT STEP into exactly one of the following categories of reasoning/behavior:

- normal operation — Straightforward, direct operations (e.g., arithmetic, factual recall, simple step-by-step logic) without introducing new logical nodes.
- deep reasoning — Multi-step causal, deductive, or analogical thinking that extends the reasoning chain by introducing new logical nodes or hidden assumptions.
- self-reflection — commenting on its own thought process (e.g., confidence, strategy, uncertainty, mistakes, or reconsideration of earlier reasoning) and tracing back to previous logical nodes.
- exploration — generating new possibilities, hypotheses, or questions, branching into alternative paths rather than following a single conclusion.

Decision rules:

(1) If multiple categories seem to overlap, choose the most specific match based on intent:
  - If the text is about reasoning itself → self-reflection.
  - If the text is branching or speculating → exploration.
  - If the text is extending the reasoning chain with deeper causality or hidden steps → deep reasoning.
  - Otherwise, if it's just direct calculation or straightforward logic → normal operation.

(2) Do not label based on correctness of the reasoning — only on the behavioral style of thinking.

(3) Ignore surface complexity (e.g., long math steps may still be normal operation if they are straightforward).

(4) If mixed, choose the dominant intent; break ties with this priority: self-reflection > exploration > deep reasoning > normal operation.

Output format (strict):
Return exactly one line and nothing else:
Behavior: {normal operation | deep reasoning | self-reflection | exploration}

PREVIOUS STEP:
{}

CURRENT STEP:
{}

Transfer Distribution Analysis To construct the reasoning topology (Figure 5), we randomly subsampled $N \in \{0.5k, 1k, 2k, 5k, 10k, 20k\}$ examples for each model-task pair. We derived reasoning transfer distributions between sequential behaviors, aggregating transition frequencies into normalized transfer graphs defined over relative step indices or semantic clusters. Stability was assessed via Pearson correlations between graphs